

Sobel kernel — Sobel edge filter on the T1 fabric (3x3 worked example)

Steps run top → bottom. Each image pixel has its own colour, so you can track its position: Step 2 moves data sideways (within a row), Step 4 moves the SAME data up/down (between rows) using the identical vslidedown / vslideup — only the mode bit changed. Gx/Gy are sign-coloured (warm +, cool −).

|                                                                                                                                                                                 | Image (v8)                                                                                                                                 | v12 right / below | v16 left / above | Gx (v20) | Gy (v24) | Output |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------|----------|----------|--------|----|----|----|----|------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|-----|----|----|-----|----|----|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|-----|-----|-----|----|----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|-----|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|----|-----------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|--|--|
| <b>1. Load image</b><br>vle8.v v8,(a0)<br>[H]                                                                                                                                   | <table><tr><td>3</td><td>6</td><td>9</td></tr><tr><td>12</td><td>15</td><td>18</td></tr><tr><td>21</td><td>24</td><td>27</td></tr></table> | 3                 | 6                | 9        | 12       | 15     | 18 | 21 | 24 | 27 | <table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>                    |    |    |    |     |    |    |     |    |    | <table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>                         |     |     |     |     |     |     |    |    |     | <table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>                      |    |    |    |    |    |     |     |     |     | <table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>                   |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
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| 12                                                                                                                                                                              | 15                                                                                                                                         | 18                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
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| <b>2. Horizontal slide</b><br>vmv.v.i v12,0; vmv.v.i v16,0<br>vslidedown.vi v12,v8,1 (right)<br>vslideup.vi v16,v8,1 (left)                                                     | <table><tr><td>3</td><td>6</td><td>9</td></tr><tr><td>12</td><td>15</td><td>18</td></tr><tr><td>21</td><td>24</td><td>27</td></tr></table> | 3                 | 6                | 9        | 12       | 15     | 18 | 21 | 24 | 27 | <table><tr><td>6</td><td>9</td><td>0</td></tr><tr><td>12</td><td>18</td><td>0</td></tr><tr><td>24</td><td>27</td><td>0</td></tr></table>       | 6  | 9  | 0  | 12  | 18 | 0  | 24  | 27 | 0  | <table><tr><td>0</td><td>3</td><td>6</td></tr><tr><td>0</td><td>12</td><td>15</td></tr><tr><td>0</td><td>21</td><td>24</td></tr></table>            | 0   | 3   | 6   | 0   | 12  | 15  | 0  | 21 | 24  | <table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>                      |    |    |    |    |    |     |     |     |     | <table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>                   |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
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| 12                                                                                                                                                                              | 15                                                                                                                                         | 18                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 21                                                                                                                                                                              | 24                                                                                                                                         | 27                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
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| 24                                                                                                                                                                              | 27                                                                                                                                         | 0                 |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
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| <b>3. Gx = right – left</b><br>vsub.vv v20,v12,v16 → Gx                                                                                                                         | <table><tr><td>3</td><td>6</td><td>9</td></tr><tr><td>12</td><td>15</td><td>18</td></tr><tr><td>21</td><td>24</td><td>27</td></tr></table> | 3                 | 6                | 9        | 12       | 15     | 18 | 21 | 24 | 27 | <table><tr><td>6</td><td>9</td><td>0</td></tr><tr><td>15</td><td>18</td><td>0</td></tr><tr><td>24</td><td>27</td><td>0</td></tr></table>       | 6  | 9  | 0  | 15  | 18 | 0  | 24  | 27 | 0  | <table><tr><td>6</td><td>6</td><td>-6</td></tr><tr><td>15</td><td>6</td><td>-15</td></tr><tr><td>24</td><td>6</td><td>-24</td></tr></table>         | 6   | 6   | -6  | 15  | 6   | -15 | 24 | 6  | -24 | <table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>                      |    |    |    |    |    |     |     |     |     | <table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>                   |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
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| 12                                                                                                                                                                              | 15                                                                                                                                         | 18                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 21                                                                                                                                                                              | 24                                                                                                                                         | 27                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 6                                                                                                                                                                               | 9                                                                                                                                          | 0                 |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 15                                                                                                                                                                              | 18                                                                                                                                         | 0                 |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 24                                                                                                                                                                              | 27                                                                                                                                         | 0                 |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 6                                                                                                                                                                               | 6                                                                                                                                          | -6                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 15                                                                                                                                                                              | 6                                                                                                                                          | -15               |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 24                                                                                                                                                                              | 6                                                                                                                                          | -24               |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                 |                                                                                                                                            |                   |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                 |                                                                                                                                            |                   |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                 |                                                                                                                                            |                   |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                 |                                                                                                                                            |                   |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                 |                                                                                                                                            |                   |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                 |                                                                                                                                            |                   |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| <b>4. Vertical slide</b><br>csrwi 0x7c0,1 (Vertical Mode)<br>vslidedown.vi v12,v8,1 (below)<br>vslideup.vi v16,v8,1 (above) [V]<br><b>flip vertical mode rotate control 90°</b> | <table><tr><td>3</td><td>6</td><td>9</td></tr><tr><td>12</td><td>15</td><td>18</td></tr><tr><td>21</td><td>24</td><td>27</td></tr></table> | 3                 | 6                | 9        | 12       | 15     | 18 | 21 | 24 | 27 | <table><tr><td>12</td><td>15</td><td>18</td></tr><tr><td>21</td><td>24</td><td>27</td></tr><tr><td>0</td><td>0</td><td>0</td></tr></table>     | 12 | 15 | 18 | 21  | 24 | 27 | 0   | 0  | 0  | <table><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>3</td><td>6</td><td>9</td></tr><tr><td>12</td><td>15</td><td>18</td></tr></table>             | 0   | 0   | 0   | 3   | 6   | 9   | 12 | 15 | 18  | <table><tr><td>6</td><td>6</td><td>-6</td></tr><tr><td>15</td><td>6</td><td>-15</td></tr><tr><td>24</td><td>6</td><td>-24</td></tr></table>      | 6  | 6  | -6 | 15 | 6  | -15 | 24  | 6   | -24 | <table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>                   |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 3                                                                                                                                                                               | 6                                                                                                                                          | 9                 |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 12                                                                                                                                                                              | 15                                                                                                                                         | 18                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 21                                                                                                                                                                              | 24                                                                                                                                         | 27                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 12                                                                                                                                                                              | 15                                                                                                                                         | 18                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 21                                                                                                                                                                              | 24                                                                                                                                         | 27                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 0                                                                                                                                                                               | 0                                                                                                                                          | 0                 |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 0                                                                                                                                                                               | 0                                                                                                                                          | 0                 |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 3                                                                                                                                                                               | 6                                                                                                                                          | 9                 |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 12                                                                                                                                                                              | 15                                                                                                                                         | 18                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 6                                                                                                                                                                               | 6                                                                                                                                          | -6                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 15                                                                                                                                                                              | 6                                                                                                                                          | -15               |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 24                                                                                                                                                                              | 6                                                                                                                                          | -24               |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                 |                                                                                                                                            |                   |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                 |                                                                                                                                            |                   |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                 |                                                                                                                                            |                   |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| <b>5. Gy = below – above</b><br>vsub.vv v24,v12,v16 → Gy<br>csrwi 0x7c0,0                                                                                                       | <table><tr><td>3</td><td>6</td><td>9</td></tr><tr><td>12</td><td>15</td><td>18</td></tr><tr><td>21</td><td>24</td><td>27</td></tr></table> | 3                 | 6                | 9        | 12       | 15     | 18 | 21 | 24 | 27 | <table><tr><td>12</td><td>15</td><td>18</td></tr><tr><td>21</td><td>24</td><td>27</td></tr><tr><td>0</td><td>0</td><td>0</td></tr></table>     | 12 | 15 | 18 | 21  | 24 | 27 | 0   | 0  | 0  | <table><tr><td>6</td><td>6</td><td>-6</td></tr><tr><td>15</td><td>6</td><td>-15</td></tr><tr><td>24</td><td>6</td><td>-24</td></tr></table>         | 6   | 6   | -6  | 15  | 6   | -15 | 24 | 6  | -24 | <table><tr><td>12</td><td>15</td><td>18</td></tr><tr><td>18</td><td>18</td><td>18</td></tr><tr><td>-12</td><td>-15</td><td>-18</td></tr></table> | 12 | 15 | 18 | 18 | 18 | 18  | -12 | -15 | -18 | <table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>                   |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 3                                                                                                                                                                               | 6                                                                                                                                          | 9                 |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 12                                                                                                                                                                              | 15                                                                                                                                         | 18                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 21                                                                                                                                                                              | 24                                                                                                                                         | 27                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 12                                                                                                                                                                              | 15                                                                                                                                         | 18                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 21                                                                                                                                                                              | 24                                                                                                                                         | 27                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 0                                                                                                                                                                               | 0                                                                                                                                          | 0                 |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 6                                                                                                                                                                               | 6                                                                                                                                          | -6                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 15                                                                                                                                                                              | 6                                                                                                                                          | -15               |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 24                                                                                                                                                                              | 6                                                                                                                                          | -24               |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 12                                                                                                                                                                              | 15                                                                                                                                         | 18                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 18                                                                                                                                                                              | 18                                                                                                                                         | 18                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| -12                                                                                                                                                                             | -15                                                                                                                                        | -18               |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                 |                                                                                                                                            |                   |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                 |                                                                                                                                            |                   |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                 |                                                                                                                                            |                   |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| <b>6.  Gx  ,  Gy </b><br>vrsb.vi v12,v20,0<br>vmax.vv v20,v20,v12<br>vrsb.vi v16,v24,0<br>vmax.vv v24,v24,v16                                                                   | <table><tr><td>3</td><td>6</td><td>9</td></tr><tr><td>12</td><td>15</td><td>18</td></tr><tr><td>21</td><td>24</td><td>27</td></tr></table> | 3                 | 6                | 9        | 12       | 15     | 18 | 21 | 24 | 27 | <table><tr><td>-6</td><td>-6</td><td>6</td></tr><tr><td>-15</td><td>-6</td><td>15</td></tr><tr><td>-24</td><td>-6</td><td>24</td></tr></table> | -6 | -6 | 6  | -15 | -6 | 15 | -24 | -6 | 24 | <table><tr><td>-12</td><td>-15</td><td>-18</td></tr><tr><td>-18</td><td>-18</td><td>-18</td></tr><tr><td>12</td><td>15</td><td>18</td></tr></table> | -12 | -15 | -18 | -18 | -18 | -18 | 12 | 15 | 18  | <table><tr><td>6</td><td>6</td><td>6</td></tr><tr><td>15</td><td>6</td><td>15</td></tr><tr><td>24</td><td>6</td><td>24</td></tr></table>         | 6  | 6  | 6  | 15 | 6  | 15  | 24  | 6   | 24  | <table><tr><td>12</td><td>15</td><td>18</td></tr><tr><td>18</td><td>18</td><td>18</td></tr><tr><td>12</td><td>15</td><td>18</td></tr></table> | 12 | 15 | 18 | 18 | 18 | 18 | 12 | 15 | 18 | <table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table> |  |  |  |  |  |  |  |  |  |
| 3                                                                                                                                                                               | 6                                                                                                                                          | 9                 |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 12                                                                                                                                                                              | 15                                                                                                                                         | 18                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 21                                                                                                                                                                              | 24                                                                                                                                         | 27                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| -6                                                                                                                                                                              | -6                                                                                                                                         | 6                 |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| -15                                                                                                                                                                             | -6                                                                                                                                         | 15                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| -24                                                                                                                                                                             | -6                                                                                                                                         | 24                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| -12                                                                                                                                                                             | -15                                                                                                                                        | -18               |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| -18                                                                                                                                                                             | -18                                                                                                                                        | -18               |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 12                                                                                                                                                                              | 15                                                                                                                                         | 18                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 6                                                                                                                                                                               | 6                                                                                                                                          | 6                 |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 15                                                                                                                                                                              | 6                                                                                                                                          | 15                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 24                                                                                                                                                                              | 6                                                                                                                                          | 24                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 12                                                                                                                                                                              | 15                                                                                                                                         | 18                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 18                                                                                                                                                                              | 18                                                                                                                                         | 18                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 12                                                                                                                                                                              | 15                                                                                                                                         | 18                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                 |                                                                                                                                            |                   |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                 |                                                                                                                                            |                   |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                 |                                                                                                                                            |                   |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| <b>7. Output = sat( Gx + Gy )</b><br>vsadd.vv v16,v20,v24<br>vse8.v v16,(a1)                                                                                                    | <table><tr><td>3</td><td>6</td><td>9</td></tr><tr><td>12</td><td>15</td><td>18</td></tr><tr><td>21</td><td>24</td><td>27</td></tr></table> | 3                 | 6                | 9        | 12       | 15     | 18 | 21 | 24 | 27 | <table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>                    |    |    |    |     |    |    |     |    |    | <table><tr><td>6</td><td>6</td><td>6</td></tr><tr><td>15</td><td>6</td><td>15</td></tr><tr><td>24</td><td>6</td><td>24</td></tr></table>            | 6   | 6   | 6   | 15  | 6   | 15  | 24 | 6  | 24  | <table><tr><td>12</td><td>15</td><td>18</td></tr><tr><td>18</td><td>18</td><td>18</td></tr><tr><td>12</td><td>15</td><td>18</td></tr></table>    | 12 | 15 | 18 | 18 | 18 | 18  | 12  | 15  | 18  | <table><tr><td>18</td><td>21</td><td>24</td></tr><tr><td>33</td><td>24</td><td>33</td></tr><tr><td>36</td><td>21</td><td>42</td></tr></table> | 18 | 21 | 24 | 33 | 24 | 33 | 36 | 21 | 42 |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 3                                                                                                                                                                               | 6                                                                                                                                          | 9                 |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 12                                                                                                                                                                              | 15                                                                                                                                         | 18                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 21                                                                                                                                                                              | 24                                                                                                                                         | 27                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                 |                                                                                                                                            |                   |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                 |                                                                                                                                            |                   |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                 |                                                                                                                                            |                   |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 6                                                                                                                                                                               | 6                                                                                                                                          | 6                 |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 15                                                                                                                                                                              | 6                                                                                                                                          | 15                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 24                                                                                                                                                                              | 6                                                                                                                                          | 24                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 12                                                                                                                                                                              | 15                                                                                                                                         | 18                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 18                                                                                                                                                                              | 18                                                                                                                                         | 18                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 12                                                                                                                                                                              | 15                                                                                                                                         | 18                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 18                                                                                                                                                                              | 21                                                                                                                                         | 24                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 33                                                                                                                                                                              | 24                                                                                                                                         | 33                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |
| 36                                                                                                                                                                              | 21                                                                                                                                         | 42                |                  |          |          |        |    |    |    |    |                                                                                                                                                |    |    |    |     |    |    |     |    |    |                                                                                                                                                     |     |     |     |     |     |     |    |    |     |                                                                                                                                                  |    |    |    |    |    |     |     |     |     |                                                                                                                                               |    |    |    |    |    |    |    |    |    |                                                                                                                             |  |  |  |  |  |  |  |  |  |